Machine Vision Course
Prof. Hamidreza Pourreza
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MiniProject-7 and 8

Texture Segmentation

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Project Abstract

Based on the provided paper, we implemented an interactive, ROI-guided texture segmentation pipeline using Laws' Texture Energy Measures **without any post-processing**. The system builds 16 Laws filters (L5, E5, S5, R5), converts them to 9 canonical energy maps via symmetric averaging, standardizes features, and assigns pixels/fragments to classes using L1 distance to ROI-derived prototypes with per-class acceptance thresholds (δ). Two operation modes are supported: **Pointwise (PW)** for fine boundaries and **Fragmental (FRAG)** for block-level robustness.

On the target images (Im501–Im503), the method **cleanly delineates texture regions** with low leakage using only the paper's core steps. PW yields crisper edges on fine patterns, while FRAG produces stable homogeneous areas on noisier textures.

Algorithm Pipeline Description

1) Input Image

We start with a grayscale image:

$$I(x, y), \quad (x, y) \in \Omega$$

where Ω is the 2D spatial domain.

2) Preprocessing (Illumination Normalization)

The image intensity is normalized by subtracting a local mean computed with a sliding window. This removes smooth lighting variations and emphasizes texture:

$$I'(x, y) = I(x, y) - \frac{1}{N^2} \sum_{u=-N/2}^{N/2} \sum_{v=-N/2}^{N/2} I(x + u, y + v).$$

3) Feature Extraction with Laws' Masks

Texture features are extracted using Laws' convolution masks.

1D kernels:

$$L5 = [1, 4, 6, 4, 1], \quad E5 = [-1, -2, 0, 2, 1], \quad S5 = [-1, 0, 2, 0, -1], \quad R5 = [1, -4, 6, -4, 1].$$

2D masks (outer product):

$$K_{ij} = v_i \otimes v_j, \quad v_i, v_j \in \{L5, E5, S5, R5\}.$$

This produces 16 filters in total.

Filter response:

$$R_{ij}(x, y) = (I' * K_{ij})(x, y).$$

Local energy (sum of absolute values in a window W):

$$E_{ij}(x, y) = \sum_{(u,v) \in W} \left| R_{ij}(x + u, y + v) \right|.$$

Algorithm Pipeline Description

4) Feature Reduction (16 $\to$ 9)

Symmetric filter pairs are averaged to avoid redundancy, and three main single features are kept:

$$E_{L5E5} = \frac{1}{2}\big(E_{L5E5} + E_{E5L5}\big), \quad \ldots$$

The final 9-dimensional feature vector at each pixel is:

$$F(x, y) = [E_1(x, y), \ldots, E_9(x, y)]^\top.$$

5) Standardization

To balance scale differences across features:

$$\tilde{F}_k(x, y) = \frac{F_k(x, y) - \mu_k}{\sigma_k}, \quad \mu_k = \text{mean}(F_k),\ \sigma_k = \text{std}(F_k).$$

6) Feature Sampling Modes

Two strategies are possible:
Pointwise mode: every pixel gets a 9D feature vector:

$$z(x, y) = \tilde{F}(x, y).$$

Fragmental mode: the image is divided into blocks of size B, and the median feature is extracted:

$$z(B) = \text{median}\{\tilde{F}(x, y) : (x, y) \in B\}.$$

7) ROI Selection and Prototype Formation

The user selects ROIs for each texture class.
Features from each ROI:

$$Z_c = \{z(x, y) : (x, y) \in R_c\}.$$

Prototype (robust representative feature vector):

$$p_c = \text{median}(Z_c, \text{axis} = 0).$$

Multiple ROIs yield multiple prototypes $\{p_c^1, p_c^2, \ldots\}$.

Algorithm Pipeline Description

8) Threshold Estimation (δ)

Distances from prototypes decide class membership.

Distance metric (L1 / Manhattan):

$$d(z, p_c) = \frac{1}{9} \sum_{k=1}^{9} \left| z_k - p_{c,k} \right|.$$

Class threshold:

$$\delta_c = \text{Percentile}_\alpha \left(\{ d(z, p_c) : z \in Z_c \} \right), \quad \alpha = 95 \text{ or } 99.$$

9) Segmentation

For each pixel/fragment, we compute distances to all class prototypes and assign the closest one if below threshold:

$$D_c(z) = \min_j d(z, p_c^j).$$

$$\hat{y}(z) = \begin{cases} \arg\min_c D_c(z), & D_c(z) < \delta_c, \\ \text{unknown}, & \text{otherwise}. \end{cases}$$

Pixels outside thresholds are marked as "unknown."

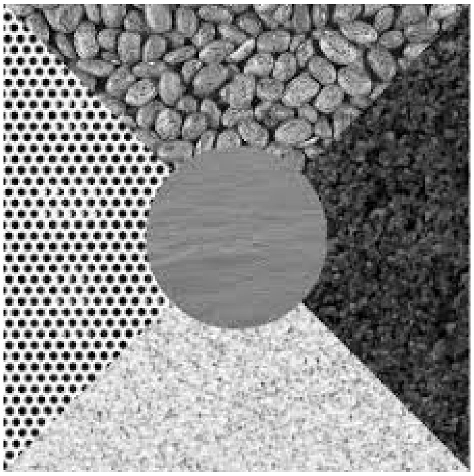
Results and Discussion

The experiments on the three test images (Im501, Im502, Im503) demonstrate that the proposed segmentation pipeline effectively identifies distinct texture regions based on Laws' energy features and ROI-based prototype selection.

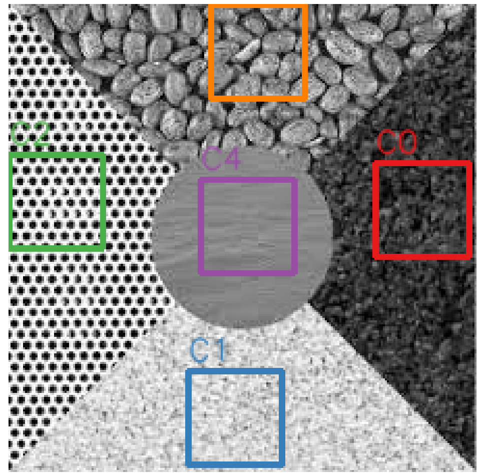
On **Im501**, the algorithm successfully captured the boundaries between multiple textures, with clear separation of the central circular region and surrounding quadrants. Both pointwise and fragmental modes produced coherent results, confirming the robustness of the method for well-structured synthetic textures. On **Im502**, the segmentation performance was more challenging due to the presence of several visually similar textures, especially on the left side of the image. This led to some local misclassifications, where boundaries between close textures became less distinct. Nevertheless, the system was still able to preserve the overall structure of the scene, correctly identifying the major regions. On **Im503**, where two large noisy regions with a curved boundary are present, multiple prototypes per class were selected. This strategy proved effective in reducing small holes and misclassified spots inside each region. The final segmentation clearly delineated the two dominant areas and produced a smooth boundary that closely follows the underlying texture transition.

Overall, the results confirm that the pipeline is capable of delivering high-quality segmentation maps, particularly in cases with well-defined textures. Even under more complex conditions with overlapping or noisy textures, the use of per-class δ thresholds and multiple prototypes helps preserve region consistency.

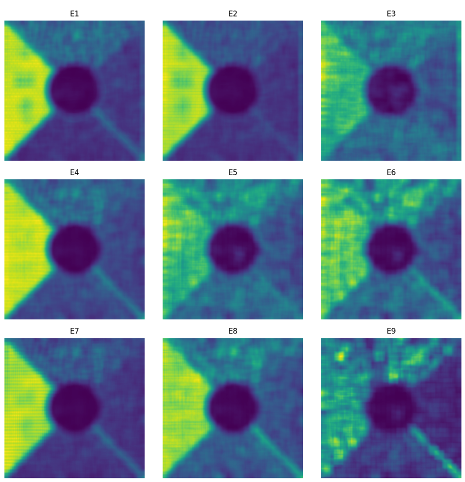
Results & Visual Analysis



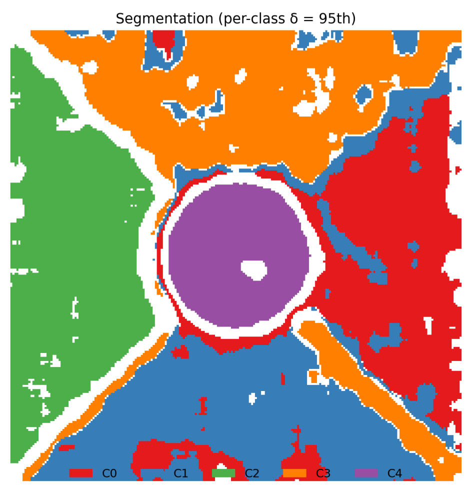
(a) Im501 (Pointwise) – Original image



(b) Im501 (Pointwise) – Selected ROIs

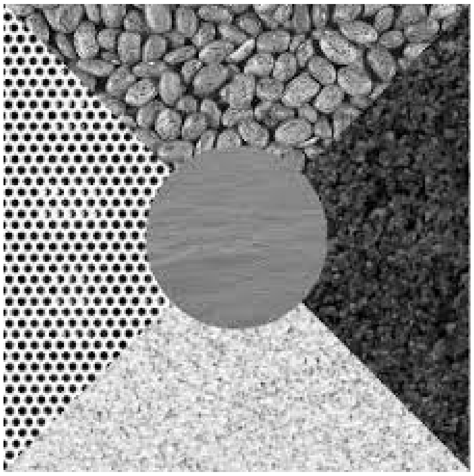


(c) Im501 (Pointwise) – Energy feature maps

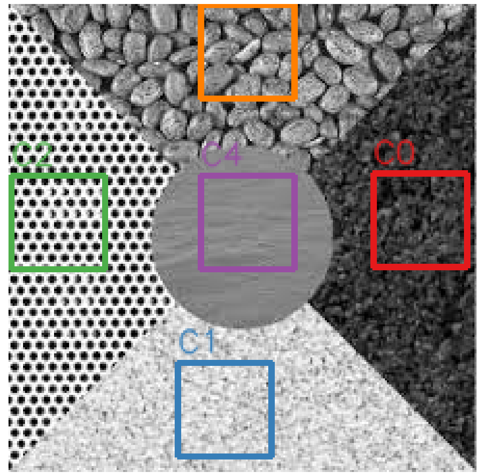


(d) Im501 (Pointwise) – Final segmentation

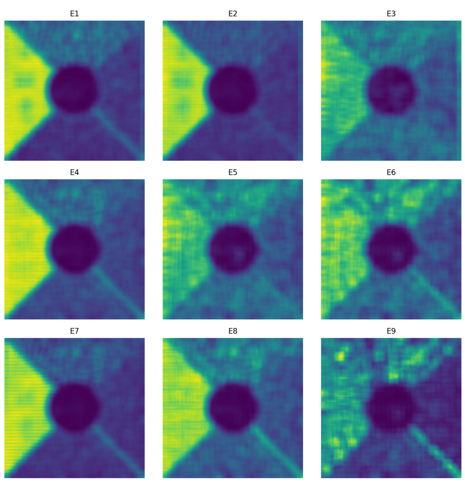
Results & Visual Analysis



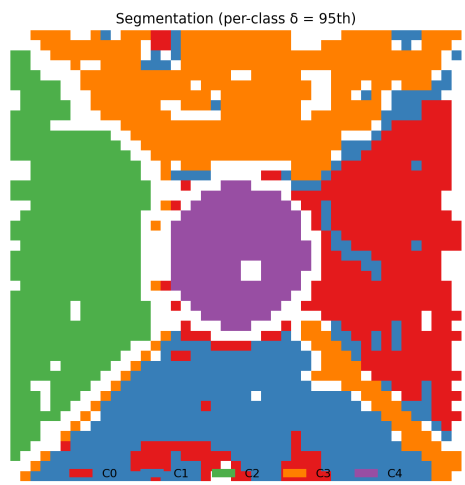
(a) Im501 (Fragmental) – Original image



(b) Im501 (Fragmental) – Selected ROIs

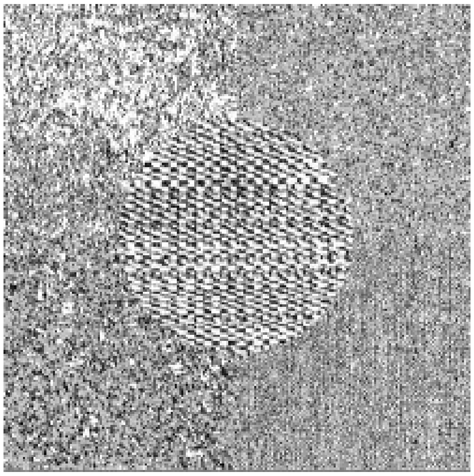


(c) Im501 (Fragmental) – Energy feature maps

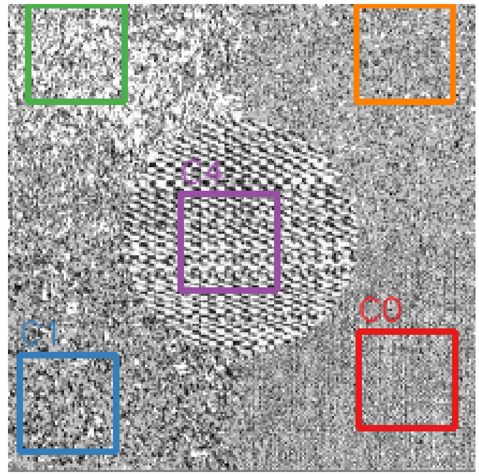


(d) Im501 (Fragmental) – Final segmentation

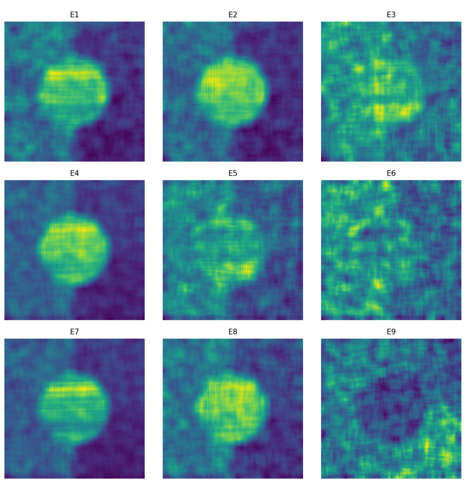
Results & Visual Analysis



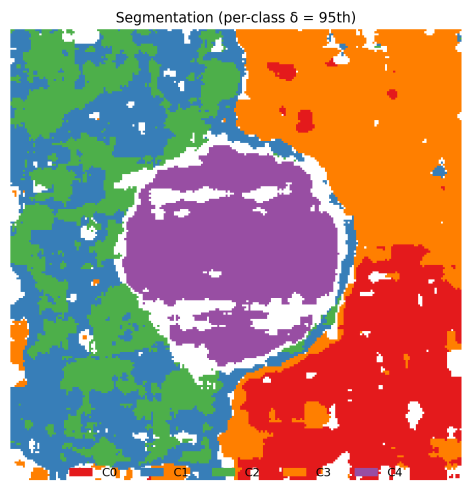
(a) Im502 (Pointwise) – Original image



(b) Im502 (Pointwise) – Selected ROIs

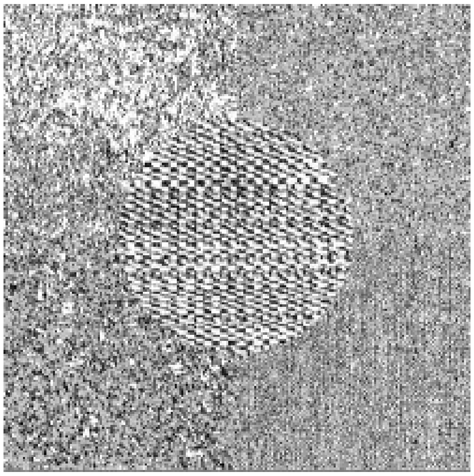


(c) Im502 (Pointwise) – Energy feature maps

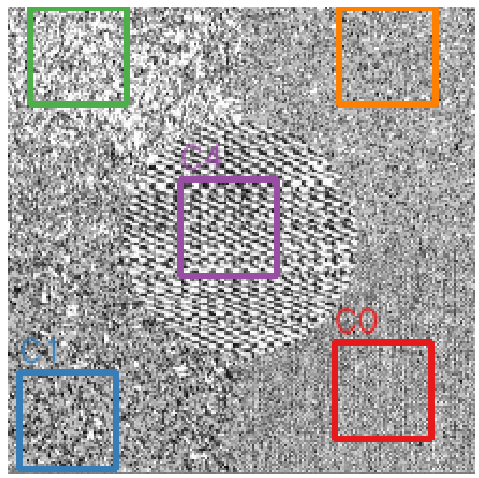


(d) Im502 (Pointwise) – Final segmentation

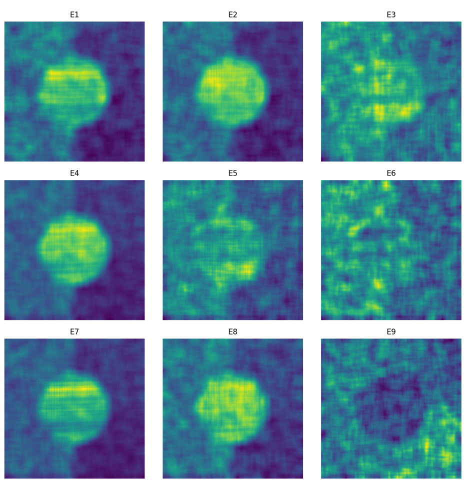
Results & Visual Analysis



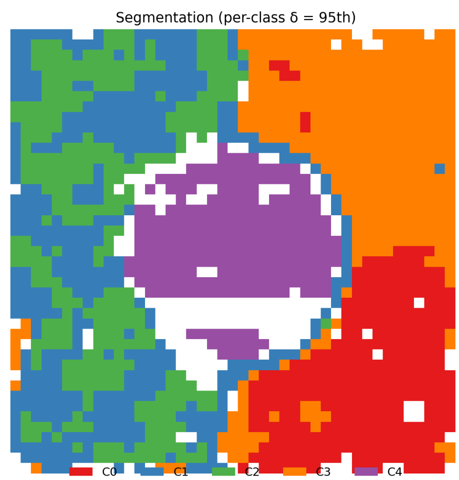
(a) Im502 (Fragmental) – Original image



(b) Im502 (Fragmental) – Selected ROIs

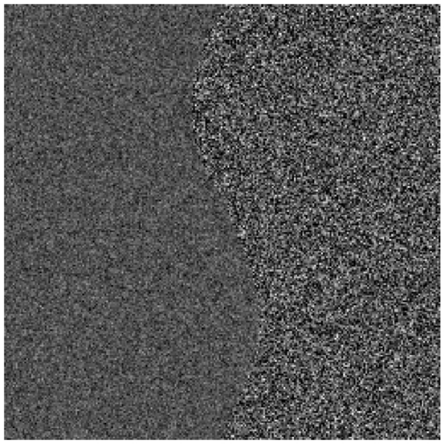


(c) Im502 (Fragmental) – Energy feature maps

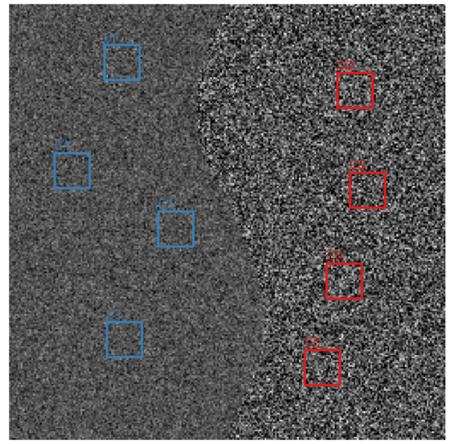


(d) Im502 (Fragmental) – Final segmentation

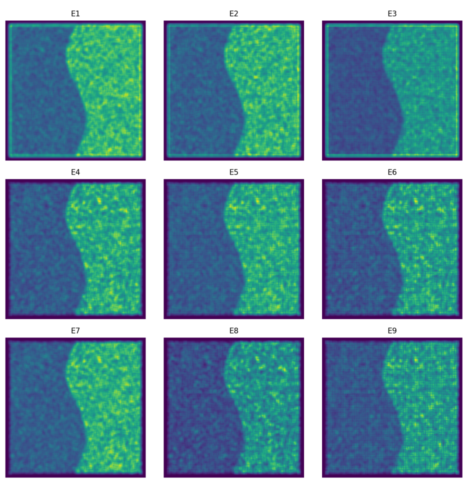
Results & Visual Analysis



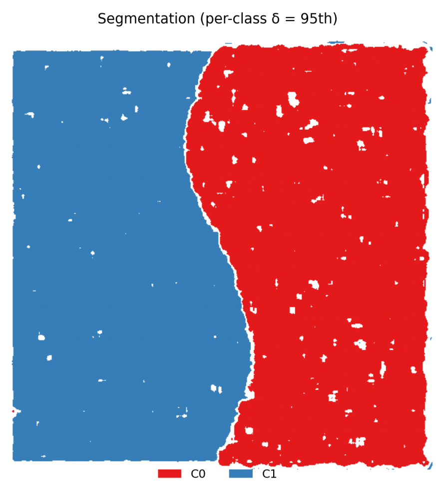
(a) Im503 (Pointwise) – Original image



(b) Im503 (Pointwise) – Selected ROIs

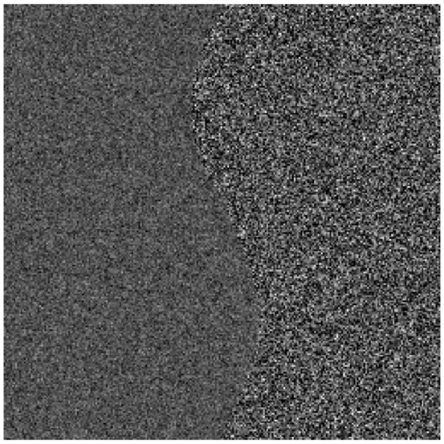


(c) Im503 (Pointwise) – Energy feature maps

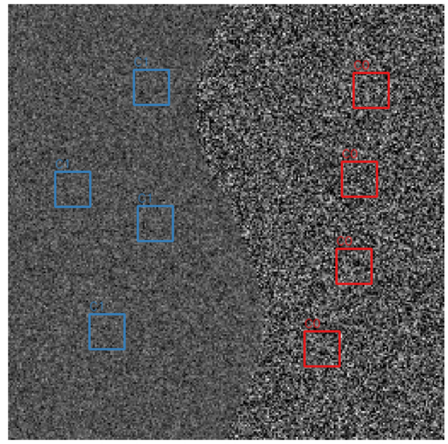


(d) Im503 (Pointwise) – Final segmentation

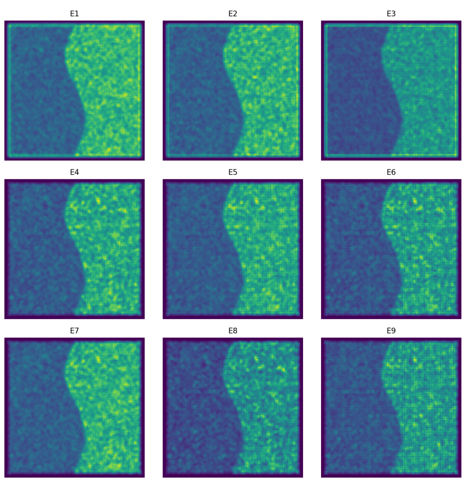
Results & Visual Analysis



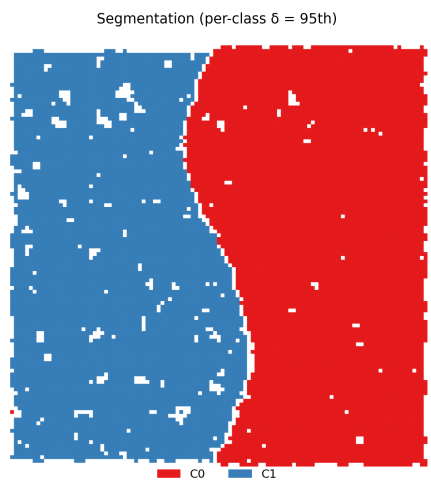
(a) Im503 (Fragmental) – Original image



(b) Im503 (Fragmental) – Selected ROIs



(c) Im503 (Fragmental) – Energy feature maps



(d) Im503 (Fragmental) – Final segmentation